

Generalized Cramer-Rao Bound for Passive MIMO Radar Multi-target Parameter Estimation

Liming Wang^{1,2}, Qian He^{1,2,*}, *Senior Member, IEEE*, and Huiyong Li^{1,2}

1. University of Electronic Science and Technology of China (UESTC), Chengdu, Sichuan 611731, China

2. Yangtze Delta Region Institute (Quzhou), UESTC, Quzhou, Zhejiang 324000, China

Abstract—Joint location and velocity estimation for multiple targets under the generalized case is studied for passive multiple-input multiple-output (MIMO) radar and the generalized Cramer-Rao bound is derived, where *generalized* means that the signals of opportunity can be nonorthogonal, the target reflection coefficients from different paths can be correlated, and the clutter-plus-noise can be spatially correlated. The results obtained can guide in implementation of the passive MIMO radar system.

I. INTRODUCTION

Multiple-input multiple-output (MIMO) radar with widely separated antenna can achieve enhanced target parameter estimation performance [1], the passive version of which [2] employs the signals of opportunity (SOO) [3] to further achieve advantages of low cost and high stealth.

Multi-target parameter estimation is a regular task in practice. Current works on MIMO radar multi-target parameter estimation rely on ideal assumptions that can hardly be satisfied in practical passive MIMO radar applications. In practice for example, different SOOs used by the passive radar may not have orthogonal baseband signals, and the radar system may suffer from multipath effect and intense interference of clutter and noise. These issues could invalidate the current analysis for the multiple targets scene. Our previous work [4] presents a Cramer-Rao bound (CRB) for general cases.

This paper investigates passive MIMO radar multi-target parameter estimation for the general case. The multi-target generalized CRB (MuGCRB) is derived to analyze the multiple targets estimation performance under the general case.

II. SIGNAL MODEL AND GENERALIZED CRAMER-RAO BOUND FOR MULTIPLE TARGET ESTIMATION

Consider an N -receiver noncoherent passive MIMO radar system employing M widely separated SOOs. In a 2-dimensional Cartesian coordinate system¹, there are Q targets located at (x^q, y^q) with velocity (v_x^q, v_y^q) . Assume that the SOOs are known to the system, then the received signal at the n th receiver and k th instant is [5] $r_n[k] = \sum_{m=1}^M \sum_{q=1}^Q \sqrt{E_m} \zeta_{nm}^q s_m(kT_s - \tau_{nm}^q) e^{j2\pi f_{nm}^q kT_s} + w_n[k]$,

*Correspondence: qianhe@uestc.edu.cn

This work was supported by the Municipal Government of Quzhou under Grant Number 2020D004.

¹Generalization to a 3-dimensional case is straightforward.

where² τ_{nm} , f_{nm} from all paths contain the target parameters $\theta = [x^1, y^1, v_x^1, v_y^1, \dots, x^Q, y^Q, v_x^Q, v_y^Q]^\top$ to be estimated. After taking K snapshots, all receivers' observation is³ $\mathbf{r} = \mathbf{S}\boldsymbol{\zeta} + \mathbf{w}$. Unlike the existing work under the ideal assumptions, our signal model does not specify any assumptions for $s_m[k]$, $\boldsymbol{\zeta}$ or $\mathbf{w}$, allowing for more generalized cases. The log-likelihood function of $\mathbf{r}$ can be computed $L(\mathbf{r}|\theta) = -\mathbf{r}^H \mathbf{C}^{-1} \mathbf{r} - \ln[\det(\mathbf{C})] - KN \ln(\pi)$, where⁴ $\mathbf{C} = \mathbf{S} \mathbf{R} \mathbf{S}^H + \mathbf{Q}$. The maximum likelihood (ML) estimate of θ is $\hat{\theta}_{ML} = \arg \max_{\theta} \{-\mathbf{r}^H \mathbf{C}^{-1} \mathbf{r} - \ln[\det(\mathbf{C})]\}$.

Based on the model above, the Fisher information matrix (FIM) can be derived⁵ as $\mathbf{J}(\theta) = \nabla_{\theta} \theta^\top \mathbf{J}(\theta) (\nabla_{\theta} \theta^\top)^\dagger$, and the MuGCRBs associated with the 2-D location and velocity estimation of the q th, $q = 1, \dots, Q$, target are⁶

$$\text{MuGCRB}_{x^q} = [(\mathbf{J}(\theta))^{-1}]_{(q-1)Q+1, (q-1)Q+1}, \quad (1a)$$

$$\text{MuGCRB}_{y^q} = [(\mathbf{J}(\theta))^{-1}]_{(q-1)Q+2, (q-1)Q+2}, \quad (1b)$$

$$\text{MuGCRB}_{v_x^q} = [(\mathbf{J}(\theta))^{-1}]_{(q-1)Q+3, (q-1)Q+3}, \quad (1c)$$

$$\text{MuGCRB}_{v_y^q} = [(\mathbf{J}(\theta))^{-1}]_{(q-1)Q+4, (q-1)Q+4}. \quad (1d)$$

²The terms ζ_{nm}^q , τ_{nm}^q , f_{nm}^q are the target reflection coefficient, time delay and Doppler frequency shift associated with the path from the m th IOO, transmitting with energy E_m , through the q th target to the n th receiver respectively, T_s is the sampling interval, $w_n[k]$ is the additive noise.

³The vector $\mathbf{r} = [r_1[1], \dots, r_N[K]]^\top$. $\mathbf{S} = \text{diag}\{\mathbf{U}_1, \dots, \mathbf{U}_N\}$, where $\text{diag}(\cdot)$ is the notation for block diagonal operation, $\mathbf{U}_n$ is a K rows matrix. The k th row of $\mathbf{U}_n$ is $[\mathbf{u}_{n1}^k, \dots, \mathbf{u}_{nM}^k]^\top$, where $(\cdot)^\dagger$ is the transpose and the block $\mathbf{u}_{nm}^k = [u_{nm}^k, \dots, u_{nm}^Q]^\top$ consists of the time delayed and Doppler shifted SOO from the m th SOO, $u_{nm}^q[k] = \sqrt{E_m} s_m(kT_s - \tau_{nm}^q) e^{j2\pi f_{nm}^q kT_s}$. $\mathbf{w} = [\mathbf{w}_1^\top, \dots, \mathbf{w}_N^\top]^\top$ and $\boldsymbol{\zeta} = [\boldsymbol{\zeta}_1^\top, \dots, \boldsymbol{\zeta}_N^\top]^\top$, composed of $\mathbf{w}_n = [w_n[T_s], \dots, w_n[KT_s]]^\top$ and $\boldsymbol{\zeta}_n = [\zeta_{n1}^\top, \dots, \zeta_{nM}^\top]^\top$ respectively, are assumed complex Gaussian $\mathbf{w} \sim \mathcal{CN}(\mathbf{0}, \mathbf{Q})$, $\boldsymbol{\zeta} \sim \mathcal{CN}(\mathbf{0}, \mathbf{R})$, where $\mathbf{0}$ is the zero column vector, $\mathbf{Q}$ and $\mathbf{R}$ are the covariance of the target reflection and noise respectively.

⁴The terms $\mathbf{C}$, $\mathbf{R}$ and $\mathbf{Q}$ are the covariance matrix of the receive observation, the target reflection coefficient from all paths and the clutter-plus-noise from all receivers respectively, $\mathbf{R}$ and $\mathbf{Q}$ are defined in last footnote.

⁵The matrix $\nabla_{\theta} \theta^\top = [\mathbf{A}_{11}, \dots, \mathbf{A}_{NM}, \mathbf{B}_{11}, \dots, \mathbf{B}_{NM}]$ is composed of $\mathbf{A}_{nm} = \text{diag}\{\mathbf{A}_{nm}^1, \dots, \mathbf{A}_{nm}^Q\}$ and $\mathbf{B}_{nm} = \text{diag}\{\mathbf{B}_{nm}^1, \dots, \mathbf{B}_{nm}^Q\}$, they can be computed as $\mathbf{A}_{nm}^q = [\partial \tau_{nm}^q / \partial x^q, \partial \tau_{nm}^q / \partial y^q, 0, 0]^\top$ and $\mathbf{B}_{nm}^q = [\partial f_{nm}^q / \partial x^q, \partial f_{nm}^q / \partial y^q, \partial f_{nm}^q / \partial v_x^q, \partial f_{nm}^q / \partial v_y^q]^\top$ respectively. $\mathbf{J}(\theta)$ is given in (2) shown in top of next page, where $\odot$ is the Hadamard product, $(\cdot)^H$ is the Hermitian transpose. Components in (2) are the covariance matrix $\mathbf{C}$, $\mathbf{R}$ and $\mathbf{Q}$. $\mathbf{Y} = \mathbf{R} \mathbf{S}^H \mathbf{C}^{-1}$, $\mathbf{S}^\top = [\mathbf{s}_1^\top, \dots, \mathbf{s}_{MNQ}^\top]$ and $\mathbf{S}^\mathbf{f} = [\mathbf{s}_1^\mathbf{f}, \dots, \mathbf{s}_{MNQ}^\mathbf{f}]$, the last two consists of $\mathbf{s}_i^\top = \mathbf{e}_n \otimes [\partial u_{nm}^q(1) / \partial \tau_{nm}^q, \dots, \partial u_{nm}^q(K) / \partial \tau_{nm}^q]^\top$ and $\mathbf{s}_i^\mathbf{f} = \mathbf{e}_n \otimes [\partial u_{nm}^q(1) / \partial f_{nm}^q, \dots, \partial u_{nm}^q(K) / \partial f_{nm}^q]^\top$ respectively, here, $\mathbf{e}_n$ denotes an $N \times 1$ vector whose n th row element is 1, $\otimes$ is the Kronecker product.

⁶The notation $[\cdot]_{p,q}$ denotes the p th row q th column element of the matrix.

$$\mathbf{J}(\boldsymbol{\theta}) = 2\Re \left\{ \begin{bmatrix} \mathbf{Y}\mathbf{S}^\tau \odot (\mathbf{Y}\mathbf{S}^\tau)^\dagger + (\mathbf{S}^\tau)^H \mathbf{C}^{-1} \mathbf{S}^\tau \odot (\mathbf{Y}\mathbf{S}\mathbf{R})^\dagger & \mathbf{Y}\mathbf{S}^f \odot (\mathbf{Y}\mathbf{S}^\tau)^\dagger + (\mathbf{S}^\tau)^H \mathbf{C}^{-1} \mathbf{S}^f \odot (\mathbf{Y}\mathbf{S}\mathbf{R})^\dagger \\ (\mathbf{Y}\mathbf{S}^f)^\dagger \odot \mathbf{Y}\mathbf{S}^\tau + (\mathbf{S}^f)^H \mathbf{C}^{-1} (\mathbf{S}^\tau)^* \odot \mathbf{Y}\mathbf{S}\mathbf{R} & \mathbf{Y}\mathbf{S}^f \odot (\mathbf{Y}\mathbf{S}^f)^\dagger + (\mathbf{S}^f)^H \mathbf{C}^{-1} \mathbf{S}^f \odot (\mathbf{Y}\mathbf{S}\mathbf{R})^\dagger \end{bmatrix} \right\} \quad (2)$$

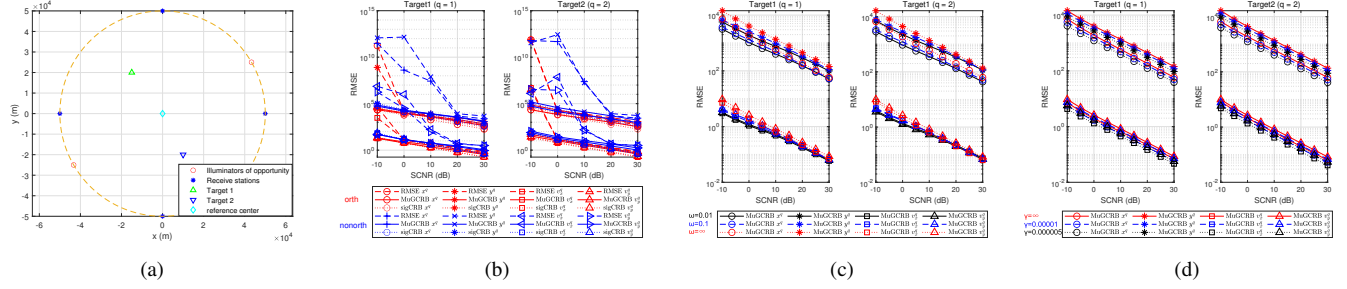


Fig. 1. (a) System setup. (b) RMSE, MuGCRB and sigCRB⁸ versus SCNR under orthogonal/nonorthogonal GMSK SOOs. (c) MuGCRB for nonorthogonal GMSK SOOs with different strength of spatial correlated target reflection coefficients, ordered by⁹ $\omega = 0.01 > \omega = 0.1 > \omega = \infty$. (d) MuGCRB for nonorthogonal GMSK SOOs with different strength of spatial correlated clutter-plus-noise, ordered by¹⁰ $\gamma = 0.000005 > \gamma = 0.00001 > \gamma = \infty$.

III. NUMERICAL EXAMPLES

Consider the scenario in Fig.1 (a) and suppose that the available SOO is the GMSK signals⁷ from the Global System for Mobile (GSM) communications system. First, we set ζ and $\mathbf{w}$ to be spatially white, Fig.1 (b) shows that the root-mean-square error (RMSE) approach the corresponding MuGCRB, this verifies the correctness of the MuGCRB. With it as a metric, Fig.1 (b) shows that $\text{MuGCRB} \approx \text{sigCRB}$ ⁸, which means that the presence of multiple targets does not greatly influence the performance, and the multiple targets estimation has performance gains from orthogonal SOOs. Second, with the spatially correlated⁹ ζ , nonorthogonal SOOs, and spatially white $\mathbf{w}$, Fig.1 (c) reveals that multiple targets estimation performance shows gains from greater spatial dependency in target reflection coefficients. Third, for the nonorthogonal SOOs, spatially white ζ and spatially colored¹⁰ $\mathbf{w}$, Fig.1 (d) shows that higher spatially correlated clutter-plus-noise have better multiple targets estimating performance. Last,

⁷The baseband waveform for the GMSK SOO is $s_m[k] = A_m \exp\{j \sum_{i=1}^{N_c} c_{mi} \sum_{l=1}^k z(lT_s - iT_p) T_s\} e^{j2\pi m \Delta f k T_s}$, where $c_{mi} \in \{-1, 1\}$ is the i -th, $i = 1, \dots, N_c$ binary data bit, N_c denotes the number of bits, $A_m = 1/\sqrt{KT_s}$ is the normalization factor, Δf is the frequency offset between SOOs, $z(t) = \pi/(2T_p) \{ \vartheta' [2\pi B/\sqrt{\ln 2}(t-T_p)] - \vartheta' [2\pi B/\sqrt{\ln 2}t] \}$, where $\vartheta'[t] = (1/\sqrt{2\pi}) \int_t^\infty e^{-\tau^2/2} d\tau$, T_p is the bit duration, and B is the 3dB bandwidth of the Gaussian prefilter used in the GMSK modulators. Set $T_p = 577\mu s$, $BT_p = 0.3577s$, let $\Delta f = 3000\text{Hz}$ for orthogonal and $\Delta f = 300\text{Hz}$ for nonorthogonal GMSK SOOs

⁸Denote the single target CRB as sigCRB.

⁹The spatially correlated ζ is modeled by its covariance matrix $\mathbf{R} = \text{diag}\{\mathbf{R}^1, \dots, \mathbf{R}^Q\}$, $\mathbf{R}^q = \mathbf{R}_t^q \otimes \mathbf{R}_r^q$ ($\otimes$ is the Kronecker product), which is composed of elements $[\mathbf{R}_r^q]_{n,n'} = \exp(-\varpi^q \Delta \phi_{r,nn'}^q)$ and $[\mathbf{R}_t^q]_{m,m'} = \exp(-\varpi^q \Delta \phi_{t,mm'}^q)$, where $\Delta \phi_{r,nn'}^q$ is the separation angle between paths from the n th and n' th receiver to the q th target, $\Delta \phi_{t,mm'}^q$ is the separation angle between paths from the m th and m' th SOO to the q th target, let $\varpi^1 = \dots = \varpi^Q = \omega$, smaller ω means stronger spatial dependencies in ζ .

¹⁰The strength of spatial correlation in $\mathbf{w}$ is controlled by γ in the elements of $\mathbf{Q} = \sigma_w^2 \tilde{\mathbf{Q}} \otimes \mathbf{I}_K$ ($\mathbf{I}_K$ is the identity matrix), which is $[\tilde{\mathbf{Q}}]_{n,n'} = \exp\{-\gamma((x_n^r - x_{n'}^r)^2 + (y_n^r - y_{n'}^r)^2)^{1/2}\}$, where σ_w^2 is the variance of $\mathbf{w}$, (x_n^r, y_n^r) is the location of the n th receiver. The smaller γ becomes, the stronger the spatial correlation is.

for the OFDM SOOs in the scenario of Fig.1 (a), we can get similar results¹¹, moreover, for other scenarios with the SOOs or number of targets of interest, the MuGCRB in (1a)-(1d) is convenient for evaluating the multi-target estimation performance.

IV. CONCLUSION

This paper derived the MuGCRB for multi-target joint location and velocity estimation using passive MIMO radar system, which allows general case with possibly nonorthogonal SOOs, correlated clutter-plus-noise, and correlated target reflection coefficients. Numerical examples have been provided to verify the correctness of the derived MuGCRB, and in the tested examples, the presence of multiple targets does not significantly influence the estimation performance, moreover, to achieve estimation performance gains for multiple targets, one can resort to choosing orthogonal SOOs, making use of spatial correlation both in the clutter-plus-noise and target reflection coefficients from different paths. In practical implementation of the passive MIMO radar system, the MuGCRB can help to benchmark the system design.

REFERENCES

- [1] H. Godrich, A. M. Haimovich, and R. S. Blum, "Target Localization Accuracy Gain in MIMO Radar-Based Systems," *IEEE Transactions on Information Theory*, vol. 56, no. 6, pp. 2783–2803, 2010.
- [2] H. D. Griffiths and C. J. Baker, "Passive coherent location radar systems. Part 1: Performance prediction," *IEE Proceedings-Radar, Sonar and Navigation*, vol. 152, no. 3, pp. 153–159, 2005.
- [3] X. Wan, J. Yi, Z. Zhao, and H. Ke, "Experimental Research for CMMB-Based Passive Radar Under a Multipath Environment," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 50, no. 1, pp. 70–85, 2014.
- [4] Q. He, J. Hu, R. S. Blum, and Y. Wu, "Generalized Cramér-Rao Bound for Joint Estimation of Target Position and Velocity for Active and Passive Radar Networks," *IEEE Trans. Signal Processing*, vol. 64, no. 8, pp. 2078–2089, 2016.
- [5] H. Godrich, A. M. Haimovich, and R. S. Blum, "Target localization accuracy and multiple target localization: Tradeoff in MIMO radars," in *2008 42nd Asilomar Conference on Signals, Systems and Computers*, 2008, pp. 614–618.

¹¹For the space limitation, numerical results of OFDM SOOs are omitted.